

# Why Local ML Models Cannot Accelerate SCF Convergence in Multiresolution Analysis: Lessons from Systematic Architecture Search

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## Abstract

We present a systematic investigation of machine learning approaches to accelerating self-consistent field (SCF) convergence in MADNESS, a multiresolution analysis (MRA) framework for density functional theory. Using a FiLM-conditioned MLP with same-level halo neighbors (MRANet, 3.0M parameters), we trained on 51 W4-11 molecules and evaluated three strategies: density coefficient prediction, parent node augmentation, and tree structure prediction. The density predictor achieved MSE ratios of  $0.41\text{--}0.98\times$  baseline at fine tree levels (10–14) but  $1.00\text{--}1.12\times$  at coarse levels (1–9), where SCF convergence is determined. Injecting the predicted density into MADNESS produced identical SCF iteration counts across all 14 molecule-threshold combinations tested. Parent node features (4.1M parameters) did not change the coarse-level ratios. A refinement classifier reached  $F1=0.860$  but increased SCF iterations  $2.6\times$ .

All three strategies hit the same limit: at coarse tree levels, the promolecular density and nuclear potential do not encode the electron-electron interactions needed for density corrections. We extract three transferable lessons for ML applied to multi-scale iterative solvers and propose Gaussian-basis densities as a richer input signal.

## 1 Introduction

MADNESS<sup>1</sup> is a multiresolution analysis (MRA) framework for density functional theory that represents functions as adaptive octree structures of polynomial scaling coefficients. The tree refines locally where the function has fine-scale features and remains coarse elsewhere, achieving systematic accuracy without a fixed basis set. Within a MADNESS DFT calculation, self-consistent field (SCF) iteration dominates wall time: each iteration rebuilds the Coulomb and exchange-correlation potentials, solves the bound-state Helmholtz equation for every orbital, and updates the density. The standard initial density guess is the promolecular density  $\rho_0$ , a superposition of isolated spherical atomic densities. Because  $\rho_0$  contains no bonding, charge transfer, or exchange-correlation information, the SCF procedure typically requires 8–15 iterations to converge. A more accurate starting density would reduce the iteration count and, proportionally, the total cost.

Machine learning models for electron density prediction have shown that neural networks can map nuclear geometry to real-space charge densities with useful accuracy.<sup>2</sup> These approaches typically predict density on fixed Cartesian grids and project the result onto whatever basis the solver requires. Working directly in the native MRA representation sidesteps grid-to-tree projection and preserves the adaptive structure that makes MADNESS efficient. The octree further decomposes the prediction problem into per-node regression tasks with local spatial context available from same-level neighbor nodes, a natural inductive bias for an MLP architecture.

We present a systematic investigation of three ML strategies for accelerating SCF convergence in MADNESS: direct prediction of converged density coefficients, parent-augmented

coefficient prediction, and tree structure (refinement) prediction. All three use a FiLM-conditioned MLP (MRANet) with same-level halo neighbor features as input. Despite strong per-node regression accuracy at fine tree levels, none of the strategies reduced SCF iteration count when the predicted density was used as an initial guess. We trace this failure to an input signal bottleneck: at the coarse tree levels that determine SCF convergence behavior, the promolecular density  $\rho_0$  and nuclear potential  $V_{\text{nuc}}$  lack the electron-electron interaction information needed to predict density corrections. The model learns an accurate identity mapping at fine levels (where  $\rho_0 \approx \rho$ ) but cannot extract the missing physics at coarse levels. From this negative result we extract three lessons transferable to ML applied to multi-scale iterative solvers: (1) input signal quality at the scale that matters for the solver sets the model ceiling, regardless of aggregate accuracy; (2) fine-level accuracy does not imply solver acceleration; and (3) multi-task interference between level-dependent objectives can silently degrade training.

## 2 Background

### 2.1 Multiresolution Analysis in MADNESS

MADNESS<sup>1</sup> represents three-dimensional functions in a discontinuous multiwavelet basis built on an adaptive octree. The simulation cell  $[0, 1)^3$  is recursively bisected along each axis to produce cubic boxes at integer levels  $n = 0, 1, 2, \dots$ , with each box at level  $n$  identified by a translation index  $\mathbf{l} \in \{0, \dots, 2^n - 1\}^3$ . Within each box, the function is expanded in a tensor product of Legendre scaling polynomials of order  $k$  (we use  $k = 8$  throughout), giving  $k^3 = 512$  scaling coefficients (s-coefficients) per node.

The connection between consecutive levels is mediated by Alpert’s two-scale relation.<sup>3</sup> A parent node’s s-coefficients can be decomposed into the s-coefficients of its eight children plus a set of wavelet difference coefficients (d-coefficients) that encode the detail lost in the coarsening. Refinement is controlled by the d-coefficient norm: if  $\|\mathbf{d}\| > \epsilon$  for a prescribed

threshold  $\epsilon$ , the node is subdivided into eight children and the representation is extended to the next level. Nodes whose d-coefficients fall below  $\epsilon$  become leaves. This adaptive refinement concentrates computational effort near nuclei and in bonding regions while keeping the representation coarse in smooth, slowly varying parts of space.

## 2.2 SCF Iteration Mechanics

Kohn-Sham DFT<sup>4</sup> reduces the many-electron problem to a set of single-particle eigenvalue equations in which the effective potential depends on the electron density  $\rho(\mathbf{r})$ . Because the potential is a functional of  $\rho$ , the equations must be solved self-consistently: an initial density defines a potential, the eigenvalue equations are solved for new orbitals, a new density is constructed, and the cycle repeats until  $\rho$  changes by less than a convergence threshold.

MADNESS uses the promolecular density  $\rho_0(\mathbf{r}) = \sum_A \rho_A^{\text{atom}}(\mathbf{r})$  as its initial guess, where  $\rho_A^{\text{atom}}$  is the spherically symmetric ground-state density of isolated atom  $A$ . This superposition is inexpensive to construct and places charge in roughly the right spatial regions, but it contains no information about chemical bonding, charge redistribution, or exchange-correlation effects. The distance between  $\rho_0$  and the converged  $\rho$  determines how many SCF iterations are needed.

MADNESS employs a multi-protocol convergence strategy.<sup>1</sup> The calculation begins at a coarse truncation threshold (typically  $\epsilon = 10^{-4}$ ), producing a shallow tree with fewer nodes. Once the density converges at this threshold, the tree is refined to a tighter threshold ( $\epsilon = 10^{-6}$  or beyond) and iteration continues. The first (coarsest) protocol dominates the total iteration count because it starts from  $\rho_0$ , the density furthest from the converged solution. Subsequent protocols start from an already-converged density at the previous threshold and typically converge in one or two iterations. Any ML model that aims to reduce SCF cost must therefore improve the density at coarse levels, where the first protocol operates and where the gap between  $\rho_0$  and  $\rho$  is largest.

## 3 Methods

### 3.1 Data Pipeline

Training data was generated from DFT calculations on 51 closed-shell molecules drawn from the W4-11 thermochemistry benchmark set,<sup>5</sup> spanning atomic species H, C, N, O, F, S, and Cl. All calculations used the LDA functional with MADNESS<sup>1</sup> multiresolution parameters  $k = 8$  (polynomial order) and  $\epsilon = 10^{-6}$  (truncation threshold), matching production precision for the adaptive tree representation.

For each molecule, three MADNESS functions were generated in a three-step pipeline. First, `dump_training_functions` produced the superposition-of-atomic-densities initial guess  $\rho_0(\mathbf{r})$  and the nuclear potential  $V_{\text{nuc}}(\mathbf{r})$  as MRA function trees serialized to HDF5. Second, `moldft` ran the SCF cycle to convergence. Third, `dump_training_functions --archive` extracted the converged density  $\rho(\mathbf{r})$  from the `moldft` checkpoint and wrote it as a third HDF5 tree. Each function is stored as a tree of  $k^3 = 512$ -element coefficient vectors (the *s-coefficients*) at each node.

**Sample construction.** The dataset builder enumerates every node in the converged  $\rho$  tree. Internal nodes (those with children) are labeled as positive samples requiring further refinement (`refine=1`); leaf nodes are positive samples that do not require refinement (`refine=0`). To provide negative examples, the builder generates the eight octree children below each leaf node by evaluating  $\rho$ 's s-coefficients at the child level via the two-scale relation (these children have vanishing wavelet coefficients by construction). The negative samples are labeled `refine=0`.

For each sample, the input features consist of (i) the  $\rho_0$  and  $V_{\text{nuc}}$  s-coefficients at the box itself (each  $k^3 = 512$  elements), (ii) the s-coefficients of  $\rho_0$  and  $V_{\text{nuc}}$  at six face-adjacent halo boxes (each  $6 \times 512$ ), and (iii) the integer tree level  $n$ . Halo s-coefficients for neighbors outside the simulation cell are zero-padded. The target is the converged  $\rho$  s-coefficient

vector at the same box. The complete HDF5 layout stores these arrays per molecule with gzip compression.

**Train/val/test split.** The 51 molecules were split by molecule: 45 for training, 3 for validation (ethanol,  $\text{SO}_2$ , HNNN), and 3 for testing ( $\text{H}_2\text{O}_2$ ,  $\text{C}_2\text{H}_2$ , glyoxal). This molecule-level split ensures that the validation and test sets probe generalization to unseen geometries and electronic structures, not merely interpolation within a known tree. The 45 training molecules produced approximately 108,000 positive samples.

### 3.2 Architecture (MRANet)

MRANet is a FiLM-conditioned multilayer perceptron with a factored halo encoder (Figure 1).

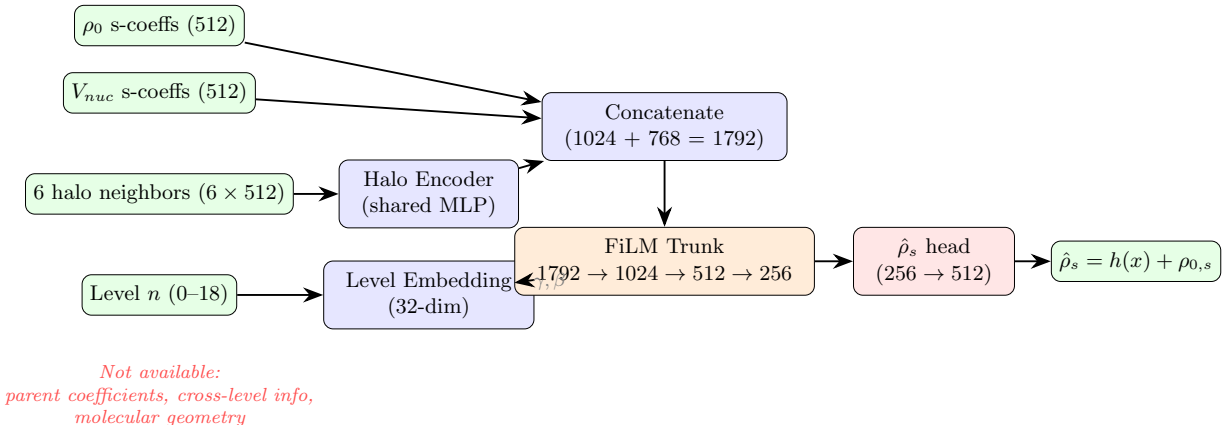


Figure 1: MRANet architecture. The model receives  $\rho_0$  and  $V_{\text{nuc}}$  s-coefficients at the center box and six face-adjacent halo boxes, conditioned on the tree level via FiLM modulation. A residual connection adds the input  $\rho_0$  coefficients to the network output.

**Halo encoder.** A shared MLP processes each of the six face-adjacent neighbor boxes identically. Each face’s input is the concatenation of the neighbor’s  $\rho_0$  s-coefficients (512),  $V_{\text{nuc}}$  s-coefficients (512), and a learnable face embedding (8-dimensional, distinguishing  $\pm x$ ,  $\pm y$ ,  $\pm z$  faces), giving an input dimension of 1032. The shared MLP maps this to a 128-

dimensional output via one hidden layer of 256 units with ReLU activation. The six per-face outputs are concatenated to produce a 768-dimensional halo representation.

**Center features.** The  $s$ -coefficients of  $\rho_0$  and  $V_{\text{nuc}}$  at the box itself are concatenated into a 1024-dimensional center feature vector.

**FiLM conditioning.** The tree level  $n \in \{0, \dots, 18\}$  indexes a 32-dimensional learned embedding table. At each trunk layer, a linear projection maps this embedding to a per-neuron scale  $\gamma$  and shift  $\beta$ , applied after batch normalization following the FiLM mechanism:<sup>6</sup>

$$\mathbf{h} = \gamma \odot \text{BN}(W\mathbf{x} + \mathbf{b}) + \beta. \quad (1)$$

This allows the network to adapt its behavior to the characteristic length scale and coefficient magnitude at each tree level without dedicating separate parameters per level.

**Trunk.** The halo representation and center features are concatenated into a 1792-dimensional input vector, which passes through three FiLM-conditioned layers with dimensions  $1792 \rightarrow 1024 \rightarrow 512 \rightarrow 256$ . Each layer applies a linear transformation, batch normalization, FiLM modulation, ReLU activation, and dropout ( $p = 0.1$ ) in sequence.

**Output head.** A single linear layer maps the 256-dimensional trunk output to a 512-dimensional vector of predicted  $\rho$   $s$ -coefficients. The output uses a residual connection to the input  $\rho_0$   $s$ -coefficients:

$$\hat{\rho}_s = f_{\theta}(\mathbf{x}) + \rho_{0,s}, \quad (2)$$

so the network learns the correction  $\rho_s - \rho_{0,s}$  rather than the full density. This inductive bias reflects the physical expectation that the converged density is close to the atomic superposition guess. The single-task model contains approximately  $3.04 \times 10^6$  parameters.

### 3.3 Training and Inference

**Loss function.** The single-task model was trained with a weighted mean squared error loss on the s-coefficient vector. To counteract the class imbalance (approximately 87% negative samples in the dataset), positive (in-tree) samples received a weight of 10 relative to negative (below-leaf) samples. The loss also incorporated level-aware masking: tree levels with fewer than 200 training samples were excluded entirely (hard cutoff), and the remaining levels were weighted by  $\sqrt{c_\ell/c_{\max}}$  where  $c_\ell$  is the sample count at level  $\ell$  and  $c_{\max}$  is the maximum count across levels. This soft weighting prevented data-sparse levels from dominating the gradient.

**Optimization.** The model was trained with AdamW (learning rate  $2 \times 10^{-4}$ , weight decay  $10^{-4}$ ) using a linear warmup over 5 epochs followed by cosine decay to a minimum learning rate of  $10^{-6}$ . Batch size was 4096 samples. Training used automatic mixed precision (AMP) with gradient scaling and ran for up to 120 epochs with early stopping (patience 20 epochs, monitored on validation loss).

**Single-task inference.** At inference time, `predict_density_simple()` copies the  $\rho_0$  tree topology (preserving internal node structure and leaf set), replaces each leaf’s s-coefficients with the model’s prediction, then runs a compress-reconstruct pass to enforce two-scale consistency throughout the tree. The predicted tree is then scaled so that  $\int \hat{\rho}(\mathbf{r}) d\mathbf{r} = N_e$ . Level clamping restricts model predictions to levels 10–14 (where training data is abundant), falling back to unmodified  $\rho_0$  coefficients at all other levels.

**SCF integration.** The predicted density was injected into MADNESS’s SCF solver as a replacement initial guess. The integration required three components: (i) a C++ modification to `SCF::initial_guess()` in `SCF.cc` (approximately 20 lines) that detects a `rhoML` archive at startup, loads it via `ParallelInputArchive`, and rescales to the correct electron count, falling back to the standard  $\rho_0$  if the archive is absent; (ii) an `h5_to_archive` con-



verter that reads the HDF5 function tree produced by Python-side inference and writes a MADNESS binary archive compatible with the C++ reader; and (iii) a driver script that runs ML inference, converts the output, and launches `moldft` with the ML-derived initial guess alongside a baseline run for comparison.

## 4 Experiments and Results

We evaluate the MRANet model in four experiments, each probing a different hypothesis about why local ML predictions might accelerate SCF convergence. Training uses 51 molecules from the W4-11 dataset<sup>5</sup> (45 train, 3 validation, 3 test) at MADNESS production settings ( $k = 8$ ,  $\epsilon = 10^{-6}$ ).

### 4.1 Density Coefficient Prediction

Our first model used multi-task training with Kendall uncertainty-weighted loss<sup>7</sup> to jointly predict density s-coefficients ( $\rho_s$ ) and refinement labels. This produced a train MSE ratio of  $2.99\times$  and validation ratio of  $2.84\times$  relative to the  $\rho_0$  baseline, meaning the model performed nearly three times *worse* than simply copying the promolecular density. Inspection of the learned uncertainty parameters revealed the cause:  $\sigma_{\rho_s}$  grew without bound during training, progressively zeroing the  $\rho_s$  gradient while the classification head consumed the entire learning signal.

Single-task ablation (removing the refinement head entirely) resolved this interference. The single-task model achieved a train ratio of  $1.006\times$  and validation ratio of  $1.004\times$ , reaching approximate parity with  $\rho_0$ .

Per-level MSE analysis on the validation molecules (ethanol,  $\text{SO}_2$ ,  $\text{HN}_3$ ) exposed a more informative pattern (Figure 2).

After applying level-aware masking (hard cutoff below 200 training samples,  $\sqrt{n/n_{\max}}$  sample weighting), the model beat  $\rho_0$  at fine levels and matched it at coarse levels:

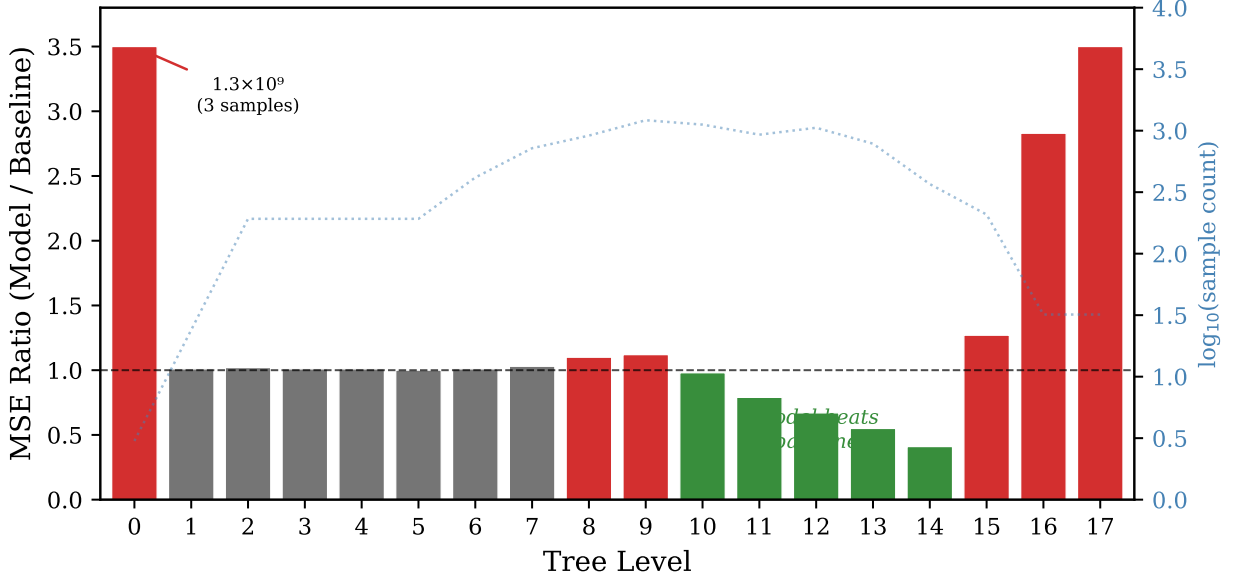


Figure 2: Per-level MSE ratio (model/baseline) on the validation set. The model beats  $\rho_0$  at levels 10–14 (ratio  $< 1$ ) but matches or slightly exceeds it at coarse levels 1–9 (ratio  $\geq 1$ ). Levels 0, 15–17 have too few training samples for reliable prediction.

- Level 0 (3 samples):  $1.34 \times 10^9 \times$  baseline. Catastrophic failure from extreme data scarcity, masked during training but exposed at inference.
- Levels 1–9 (24–1216 samples):  $1.00$ – $1.12 \times$ . Near-parity with  $\rho_0$ .
- Level 10 (1120 samples):  $0.98 \times$ .
- Level 11 (928 samples):  $0.79 \times$ .
- Level 12 (1056 samples):  $0.67 \times$ .
- Level 13 (784 samples):  $0.55 \times$ .
- Level 14 (368 samples):  $0.41 \times$ .
- Level 15 (208 samples):  $1.27 \times$ . Performance degrades as sample counts drop below the masking threshold.
- Levels 16–17 (32 samples each):  $2.83 \times$  and  $5.34 \times$ .

The model learns a genuine correction to  $\rho_0$  at levels 10–14, where training data is abundant and the density correction signal is spatially localized near nuclei. At coarser levels, the correction is a non-local function of electron-electron interactions that cannot be inferred from same-level halo features alone.

We constructed a level-clamped predictor that uses the ML model at levels 10–14 and falls back to  $\rho_0$  elsewhere. This hybrid achieved an aggregate validation MSE of  $0.97\times$  baseline: a 3% improvement over  $\rho_0$ . The natural question is whether this improvement translates to fewer SCF iterations.

## 4.2 Parent Node Features

One hypothesis for the coarse-level plateau is that the model lacks multi-resolution context. To test this, we concatenated parent-node  $\rho_0$  and  $V_{\text{nuc}}$  s-coefficients (1024 additional floats per sample) to the trunk input, increasing it from 1792 to 2816 dimensions and total parameters from 3.0M to 4.1M. Because the first layer shape changed, we trained from scratch rather than fine-tuning.

The result was negative. Per-level MSE ratios were identical to those of the model without parent features, within training noise. The parent  $\rho_0$  and  $V_{\text{nuc}}$  encode the same atomic superposition signal at coarser resolution. They do not contain the density correction (exchange-correlation, orbital relaxation, bonding redistribution) because that correction is what the converged SCF procedure computes. Adding a coarser copy of the input does not introduce new information about the target.

## 4.3 SCF Convergence Test

We injected the level-clamped density into MADNESS and ran full SCF calculations on 7 molecules at two convergence thresholds ( $10^{-6}$  and  $10^{-8}$ ), yielding 14 comparisons against the  $\rho_0$  baseline (Table 1).

All 14 comparisons produced identical iteration counts.  $\text{CH}_3\text{OH}$  converged in 12 iterations

Table 1: SCF iteration counts for level-clamped ML density (levels 10–14) vs. promolecular baseline ( $\rho_0$ ). All 14 comparisons yield identical iteration counts. Ground-state energies agree to 8+ significant figures.

| Molecule                      | $N_e$ | Thresh    | Baseline | ML | $E_{\text{base}}$ (Ha) |
|-------------------------------|-------|-----------|----------|----|------------------------|
| CH <sub>3</sub> OH            | 18    | $10^{-6}$ | 12       | 12 | -114.85038034          |
| CH <sub>3</sub> OH            | 18    | $10^{-8}$ | 12       | 12 | -114.85038034          |
| Ethanol                       | 26    | $10^{-6}$ | 12       | 12 | -153.81559584          |
| Ethanol                       | 26    | $10^{-8}$ | 12       | 12 | -153.81559584          |
| SO <sub>2</sub>               | 32    | $10^{-6}$ | 14       | 14 | -546.34469275          |
| SO <sub>2</sub>               | 32    | $10^{-8}$ | 14       | 14 | -546.34469265          |
| HN <sub>3</sub>               | 22    | $10^{-6}$ | 13       | 13 | -163.57199524          |
| HN <sub>3</sub>               | 22    | $10^{-8}$ | 13       | 13 | -163.57199524          |
| H <sub>2</sub> O <sub>2</sub> | 18    | $10^{-6}$ | 12       | 12 | -150.55376660          |
| H <sub>2</sub> O <sub>2</sub> | 18    | $10^{-8}$ | 12       | 12 | -150.55376660          |
| C <sub>2</sub> H <sub>2</sub> | 14    | $10^{-6}$ | 10       | 10 | -76.63064903           |
| C <sub>2</sub> H <sub>2</sub> | 14    | $10^{-8}$ | 10       | 10 | -76.63064903           |
| Glyoxal                       | 30    | $10^{-6}$ | 13       | 13 | -226.15979738          |
| Glyoxal                       | 30    | $10^{-8}$ | 13       | 13 | -226.15979738          |

with both initial guesses. Ethanol: 12. SO<sub>2</sub>: 14. HN<sub>3</sub>: 13. H<sub>2</sub>O<sub>2</sub>: 12. C<sub>2</sub>H<sub>2</sub>: 10. Glyoxal: 13. Ground-state energies agreed to 8 or more significant figures (e.g.,  $E_{\text{CH}_3\text{OH}} = -114.85038034$  Ha for the baseline vs.  $-114.85038032$  Ha for the ML density). The correct electronic state was recovered in every case.

The mechanism is straightforward. MADNESS uses a multi-protocol SCF strategy: the first protocol solves at a coarse threshold ( $\epsilon = 10^{-4}$ ), and subsequent protocols refine progressively. The coarse protocol sets the density at levels 1–9, which determines the convergence path. Fine-level improvements at levels 10–14, where the ML model outperforms  $\rho_0$ , are projected away when the tree is compressed to the coarse threshold. By the time later protocols operate at fine resolution, the density has already been updated by SCF iterations and the initial guess is irrelevant.

Tightening the final threshold from  $10^{-6}$  to  $10^{-8}$  does not change this conclusion. The additional refinement occurs at even finer levels, but the coarse-level convergence path remains identical.

## 4.4 Tree Structure Prediction

The density prediction results suggest that the local receptive field limits regression accuracy at coarse levels. We tested whether the same limitation applies to a simpler task: predicting which nodes require adaptive refinement (a binary classification problem).

Multi-task training (density regression + refinement classification) achieved  $F1 = 0.857$ . The learned  $\sigma_{\rho_s}$  shrank to 0.075, upweighting the density loss by  $89\times$  and starving the refinement classification head. Training the refinement head alone with focal loss<sup>8</sup> ( $\gamma = 2$ ,  $\alpha = 0.75$ ) yielded  $F1 = 0.860$ , a delta of 0.003. Removing multi-task interference had negligible effect on the classification ceiling.

We tested the refine-only model in an SCF tree-walk: starting from the ML-predicted tree structure rather than the  $\rho_0$  tree.  $\text{CH}_3\text{OH}$  required 31 iterations compared to 12 with the baseline tree, a factor of  $2.6\times$  worse. The model did recover the correct electronic state (energies agreed to 7 significant figures:  $-114.85038034$  vs.  $-114.85037863$  Ha), confirming that the SCF procedure corrects the tree errors at the cost of additional iterations.

The classification task faces the same receptive field limitation as regression. Whether a node at level  $N$  should be refined depends on the density structure at levels 0 through  $N - 1$ , which determines the wavelet coefficients that trigger refinement in MADNESS. A model restricted to same-level halo neighbors cannot access this information.

## 5 Analysis

The experiments in Section 4 present a consistent picture: the model learns a genuine correction to  $\rho_0$  at fine tree levels (10–14) but cannot improve on the promolecular density at coarse levels (1–9). Since the coarse levels determine SCF convergence, the fine-level improvement has no effect on iteration count. This section explains why.

MADNESS solves the Kohn–Sham equations<sup>4</sup> through a multi-protocol SCF strategy.<sup>1</sup> The first protocol operates at a coarse truncation threshold ( $\epsilon = 10^{-4}$ ), which resolves only

tree levels 1 through approximately 9. At this threshold, the adaptive tree discards all structure below the coarse truncation scale: wavelet coefficients at levels 10–14 fall below  $\epsilon$  and are projected away when the Fock matrix is constructed. The SCF cycle at this coarse protocol determines the number of iterations required and the electronic state reached. Once the coarse protocol converges, subsequent protocols inherit the converged coarse density and refine it by adding spatial detail at progressively finer levels. These later protocols do not revisit the coarse-level convergence path. They begin from a density that is already self-consistent at the coarse scale and simply resolve additional spatial structure (Figure 3).

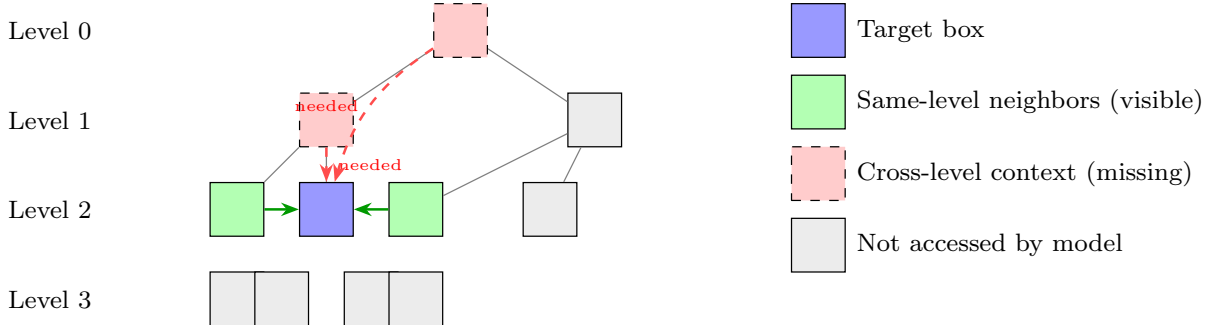


Figure 3: Receptive field limitation of the halo-neighbor architecture. The model at a target node (blue) sees only same-level face-adjacent neighbors (green). Cross-level context from parent and root nodes (red, dashed) is needed to capture the global electronic structure that determines the density correction at coarse levels.

This multi-protocol structure explains the SCF test results directly. The level-clamped predictor improves on  $\rho_0$  only at levels 10–14. When MADNESS compresses the initial guess to the first-protocol threshold, these improvements vanish. The coarse-level density seen by the first SCF protocol is identical whether the initial guess is  $\rho_0$  or the ML-corrected density. The iteration counts match because the two initial guesses are, at the resolution that matters, indistinguishable.

The question then becomes: why does the model fail at coarse levels? The answer lies in what the input features encode and what the target correction requires. At any tree level,  $\rho_0$  is a smooth superposition of spherically symmetric atomic densities.  $V_{\text{nuc}}$  is the nuclear Coulomb potential. Both are functions of atomic positions and species alone. The target

correction  $\Delta\rho = \rho_{\text{conv}} - \rho_0$  encodes exchange-correlation effects, orbital relaxation, and the redistribution of electron density due to chemical bonding. These are consequences of the full many-electron wavefunction. They arise from non-local electron-electron interactions that the self-consistent field procedure computes iteratively by solving the Kohn–Sham eigenvalue problem over the entire molecular domain.

At fine levels (10–14), the correction  $\Delta\rho$  is spatially localized near nuclei and varies smoothly as a function of the local atomic environment. Same-level halo neighbors provide enough context for the model to interpolate this correction from training data. At coarse levels,  $\Delta\rho$  reflects global electronic structure: how orbitals delocalize across the molecule, how exchange-correlation modifies the density at scales comparable to bond lengths, and how the occupied orbital space responds to the full molecular potential. The input features at a coarse node, consisting of  $\rho_0$  and  $V_{\text{nuc}}$  at that node and its six face-adjacent neighbors, do not contain this information and cannot contain it. The promolecular density and nuclear potential at a handful of neighboring boxes encode atomic positions and charges, not the self-consistent electronic response to those charges.

This is not a model capacity problem. The parent feature experiment (Section 4.2) increased parameters from 3.0M to 4.1M with no change in coarse-level accuracy. It is not a data problem. Expanding the training set from 15 to 51 molecules improved fine-level predictions but left coarse-level ratios unchanged at 1.00–1.12 $\times$  baseline. It is an input sufficiency problem: the mapping from  $(\rho_0, V_{\text{nuc}})$  to  $\Delta\rho$  at coarse levels is not a well-defined function of those inputs. Multiple distinct corrections  $\Delta\rho$  can arise from the same local  $\rho_0$  and  $V_{\text{nuc}}$  pattern, because the correction depends on the global molecular context that the local inputs do not resolve.

Could a different architecture overcome this limit? The parent feature experiment tested the simplest form of multi-scale context: concatenating the  $\rho_0$  and  $V_{\text{nuc}}$  coefficients from the parent node. This expanded the receptive field to span two tree levels, but the additional information was a coarser copy of the same signal. A graph neural network or attention-

based architecture that aggregates information across the entire tree would have access to every node in the adaptive representation. However, if the node-level features remain  $\rho_0$  and  $V_{\text{nuc}}$ , then global aggregation produces a summary of the promolecular density and nuclear potential, not of the self-consistent density. The quantity the model needs to predict, the electron-electron interaction correction, is precisely what is absent from these inputs. No aggregation scheme over  $\rho_0$  and  $V_{\text{nuc}}$  can reconstruct  $\Delta\rho$  at coarse levels, because  $\Delta\rho$  is not a function of  $\rho_0$  and  $V_{\text{nuc}}$  alone. It depends on the solution of the Kohn–Sham equations, which is the output of the SCF procedure, not a property of the input potential.

This is the fundamental limit. A local ML model that takes the promolecular density and nuclear potential as input and produces a density correction as output is attempting to bypass the SCF iteration by learning its fixed point from input data that does not determine that fixed point. The model succeeds at fine levels because there the fixed-point correction happens to be a smooth, spatially localized function that correlates with the local atomic environment. It fails at coarse levels because there the correction is a global property of the electronic state. The multi-protocol SCF structure of MADNESS then ensures that only the coarse-level result affects convergence speed. The fine-level success, while real, is operationally irrelevant.

## 6 Lessons and Path Forward

Three observations from this work generalize beyond the MRA-NN setting and may inform future attempts to couple ML models with iterative scientific solvers.

Input signal quality determines the ceiling of any downstream model, independent of architecture capacity. Our parent model doubled the input dimension (2816 vs. 1792 features) and added 34% more parameters (4.1M vs. 3.0M), yet reproduced the per-level MSE ratios of the base model almost exactly. The single-task ablation confirmed that multi-task interference was a real training pathology, but removing it only brought the model to parity with



$\rho_0$ , not beyond it. The fundamental limitation is that the promolecular density and nuclear potential carry no electron-electron correlation signal. No amount of nonlinear capacity can extract exchange-correlation physics from inputs that do not encode it.

Fine-scale accuracy does not imply solver acceleration. The level-clamped model achieved MSE ratios of  $0.41\text{--}0.98\times$  baseline at levels 10–14, a measurable and statistically significant improvement over  $\rho_0$ . Yet SCF injection tests produced identical iteration counts across all 14 molecule-threshold combinations (7 molecules, 2 thresholds). The explanation is structural: MADNESS protocols<sup>1</sup> refine from coarse to fine, and iteration count is determined by the coarsest protocol stage where the initial density deviates from the converged solution. Improvements at spatial scales finer than the first protocol threshold provide no leverage on convergence. Any ML model targeting solver acceleration must improve the density at the resolution the solver actually queries first, not at the resolution where data is most abundant.

Multi-task interference in uncertainty-weighted losses can silently undermine the primary objective. Kendall uncertainty weighting<sup>7</sup> allowed the learned scale parameter  $\sigma_{\rho_s}$  to grow without bound, progressively zeroing the density gradient while the classification head consumed the entire learning signal. The resulting model was  $2.99\times$  worse than  $\rho_0$  on density prediction, yet the aggregate loss decreased monotonically, masking the failure. A subsequent configuration flag (`refine_focused`) appeared to fix this by switching the checkpoint selection metric to refinement F1, but it left the loss function unchanged; the density head still trained under the corrupted gradient. Refine-only training with pure focal loss achieved  $F1=0.860$ , a gain of only 0.003 over the multi-task formulation ( $F1=0.857$ ). The broader warning is that multi-task losses with learned weighting parameters require per-head monitoring of both loss and scale; aggregate convergence is not a reliable diagnostic.

## 6.1 Path Forward: Gaussian-Basis Density as ML Input

The negative results above point to a concrete next step. If the bottleneck is input signal quality, the remedy is to supply an input that already contains exchange-correlation physics.

A natural candidate is the converged density from a Gaussian-basis DFT calculation (e.g., Dalton), projected onto the MRA grid. Preliminary experiments by collaborators show that using the Dalton density directly as the MADNESS starting guess reduces SCF iteration counts, confirming that the missing signal is the core issue.

This reframes the ML problem from predicting a large correction (promolecular  $\rho_0 \rightarrow$  converged MRA  $\rho$ ) to predicting a small residual (Gaussian-basis  $\rho_{\text{GTO}} \rightarrow$  converged MRA  $\rho$ ). The residual is dominated by basis-set truncation artifacts near nuclei and in diffuse tails, both of which are spatially localized and well-suited to the local halo-feature architecture already developed. We plan to construct training data by pairing Dalton densities with converged MADNESS densities for the same molecules and to retrain the single-task model on the resulting residuals. No results from this approach are available yet, and the magnitude of the achievable speedup remains an open question.

## 7 Conclusion

We trained three local ML models to accelerate SCF convergence in MADNESS and all three failed. A level-clamped density predictor achieved MSE ratios of  $0.41\text{--}0.98\times$  baseline at fine tree levels (10–14) and  $1.00\text{--}1.12\times$  at coarse levels (1–9), producing identical SCF iteration counts across all 14 molecule-threshold combinations. A parent-node feature model expanded the architecture to 4.1M parameters and doubled the input dimension, yet reproduced the same per-level MSE ratios as the base model, confirming that additional capacity extracts no new signal from the available inputs. A tree structure predictor reached  $F1 = 0.860$  but required 31 SCF iterations versus 12 for the promolecular guess, a  $2.6\times$  degradation.

The common root cause is an input signal bottleneck at coarse tree levels. MADNESS converges its SCF solution at levels 1–9 before refining finer scales, so only coarse-level density corrections can reduce iteration counts. The promolecular density and nuclear potential at those levels encode atomic superposition and electrostatics but carry no electron-electron

interaction information. No local model operating on these inputs can produce the exchange-correlation corrections needed to improve the coarse-level density, regardless of architecture or training procedure.

This mechanistic explanation (that multi-scale solvers gate convergence at their coarsest resolution, where atom-centered inputs lack the relevant physics) is not specific to MADNESS or to density functional theory. It applies whenever an ML model receives only single-body inputs and targets a quantity governed by many-body interactions, and the iterative solver queries coarse scales first. Recognizing this pattern before training avoids the cycle of architecture search and hyperparameter tuning that preceded our diagnosis.

One path remains open. A converged Gaussian-basis density already contains exchange-correlation physics and, when used directly as a MADNESS starting guess, reduces SCF iteration counts. An ML model trained on the small residual between the Gaussian-basis density and the converged MRA density would operate on an input that does not suffer from the signal bottleneck identified here. Whether this residual is learnable with local features, and whether the resulting speedup justifies the cost of the Gaussian-basis calculation, are questions that require new experiments.

## Acknowledgement

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